

Jack Hopkins

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US and Canadian citizen — authorized to work in the US without sponsorship; open to relocation

SUMMARY

Technical Gameplay Designer at World Makers (Cambridge, UK), combining hands-on Unreal Engine 5 implementation with design ownership. Shipped Deceit 2, a live-service social deduction horror game, leading DLC releases and owning multiple maps from prototype through to live release.

Currently building MVX: London, a massive-scale PvE shooter, owning the “physicality” system, a dynamic physics framework linking characters, creatures and hero abilities.

Holds a First Class MEng in Computing (Games Engineering) from Newcastle University, with a C++ foundation in real-time physics simulation, collision systems and OpenGL rendering. Co-founded and led art direction for Astral Dawn Studios, an indie game studio built with two university friends, and has produced video content, including a personal YouTube channel, since 2009.

EXPERTISE

- **Unreal Engine 5 Production:** C++; Gameplay Ability System (GAS); networked Blueprint; UMG widgets; animation and MetaHuman pipelines; Control Rig; LOD/HLOD.
- **Technical Design:** Prototyping, feature ownership, full design-to-release pipeline
- **Production Pipeline and Delivery:** Perforce (P4V) and Unreal Game Sync (UGS) for version control; TeamCity and Horde for build automation; Git for personal and smaller-scale projects; Jira for sprint planning and issue triage.
- **Languages and Engines:** Proficient in C++, C#, Java, Unity, HTML and CSS; working knowledge of C, Godot, Assembly, JavaScript and MySQL.
- **Cross-Discipline Leadership:** Communicating design vision across art, engineering and production, including coordination of local and international outsourcing teams
- **Player Research & Feedback:** Playtesting coordination, community feedback loops, data-informed iteration
- **Art and Content Tools:** Character and environment pixel art, tile sets and battle animation; 3D modelling in Blender; music composition for game environments and combat; video editing for YouTube-format content.

PROFESSIONAL EXPERIENCE

Technical Gameplay Designer — *World Makers, Cambridge, UK* Jan 2025 – Present

- Own feature delivery on Deceit 2 and MVX: London in Unreal Engine 5, leading DLC releases and owning multiple maps from concept through to live release.
- Design and implement “physicality,” a dynamic physics framework linking character, creature and hero-ability interactions, as a core system of MVX: London.
- Build gameplay systems using GAS architecture, networked Blueprint and UMG widgets, engineering for reusability so the wider team can iterate without a programmer in the loop.
- Operate the full production pipeline (Perforce, Unreal Game Sync, TeamCity, Horde, Jira) to keep builds stable and features shipping on schedule.
- Coordinate cross-discipline delivery across art, engineering and production, including local and international outsourcing teams.
- Present feature and roadmap updates directly to players in both live and scripted formats.

Graduate Gameplay Programmer — *World Makers, Cambridge, UK* Oct 2022 – Jan 2025

- Implemented gameplay on Deceit 2 across architecture, tooling and profiling, working extensively with GAS, Blueprint and UE5’s animation and UMG systems.
- Took designer-authored features from prototype through to polished, shipped releases across multiple live updates.
- Built Unreal Engine level blockouts across disciplines, aligning creative teams around a shared technical plan.
- Gathered and triaged feedback from long-term players and playtesters, translating it into concrete engineering and design changes.

EDUCATION

MEng Computing (Games Engineering) — *Newcastle University, Newcastle, UK* Sep 2018 – Jun 2022

- Grade: First Class with Honours.
- Core modules across C++ (real-time 2D/3D physics simulation, OpenGL rendering pipeline, networking), C#, cryptography, concurrency, and bioinformatics and biologically-inspired computer algorithms.

HOBBY EXPERIENCE - ASTRAL DAWN STUDIOS

Co-Founder and Lead Artist — *Astral Dawn Studios, Windsor, UK* Jun 2019 – Oct 2020

- Co-founded an indie game studio with two university friends (Karsten Finlay and Lars Finlay), self-directing development in Unity from concept to a public demo release.
- Led art direction and asset creation across two projects, an early JRPG (Astral Dawn) and a 2D platformer (Dodo Alone), producing character portraits, tile sets, battle animations and 3D character models built in Blender.
- Composed and produced original music tracks for game environments and combat encounters.
- Produced devlogs, tutorials and a podcast on dialogue and character writing in games, building a community following around the studio's development process.

SELECTED ACADEMIC AND TECHNICAL PROJECTS

Newcastle University Coursework

- **Procedural Animation System** (*Master's Dissertation, Unreal Engine, Blueprint/Control Rig, 2022*): implemented automated limb placement, ragdoll and physical animation blending; researched rigid-body animation techniques including Featherstone's algorithm (88%).
- **Automatic City Generator** (*Bachelor's Dissertation, C#/Unity, 2021*): built a settlement generator driven by a colour-coded layout diagram for fast, quality-preserving city creation.
- **Guided Missile System** (*Unreal Engine, C++, 2022*): implemented a full guided-missile targeting and control system, written entirely in C++, as a technical assessment.
- **Tropical Island** (*Advanced Graphics for Games, C++/OpenGL, 2021*): implemented a skybox, normal and height maps, and an automated moving camera.
- **Screwball Scramble** (*Advanced Game Technologies, C++, 2021*): built collision detection and resolution, menus, SFX and state-based AI on top of a custom OpenGL render setup.
- **An Afternoon with the Queen** (*Game Development, C#/Godot, 2021*): built a multi-board variant of noughts and crosses from scratch, including a minimax and alpha-beta-pruning AI opponent.
- **Collatz Password Manager** (*Advanced Programming for Games, C++, 2020*): built a password manager using the Collatz function for encryption, authenticating and testing password strength.
- **Sword Island** (*Graphics for Games, Unity, 2020*): implemented planetary textures, environmental lighting, water shaders and a day/night cycle.
- **Game Simulations Project** (*C++, Visual Studio, 2020*): added Newtonian physics, collision detection and response, and basic AI pathfinding to an established codebase.

Game Jams and Personal Projects

- **GMTK Game Jam 2021** (*48-hour team jam, Jun 2021*): built a game with three university friends on the theme "Joined Together"; ranked 860th of 5,800+ entries, 423rd for Presentation, which Jack led.
- **Ludum Dare 44: Old Macdonald's Farm** (*72-hour team jam, 2019*): built a tower-defence game with two friends on the theme "Life is Currency."

VIDEO AND CONTENT CREATION

- **Jarkey Bacon** (personal YouTube channel, since 2009): 500,000+ views and 500+ subscribers; writing, editing and audience engagement across commentary and analysis content.
- **KyleEntertainment** (freelance editing): edited manga music videos and narrative video essays for a media analysis channel.